

## FIELDNODE · DESIGN DECISIONS REGISTER

# FieldNode design decisions register

Design Molecule Lab · designmolecule.com

## DOCUMENT CONTROL

|             |                                           |         |                |
|-------------|-------------------------------------------|---------|----------------|
| Document ID | FND-DEC-001                               | Version | 0.3            |
| Status      | <b>Draft</b>                              | Date    | 2026-10-02     |
| Author      | Amish Chadha                              | License | CERN-OHL-S-2.0 |
| Project TRL | <b>TRL 3</b> Proof of concept             |         |                |
| Repository  | github.com/BoujeeEnjinia1701/fieldnode    |         |                |
| Source      | docs/06-design-decisions.md @ uncommitted |         |                |

## REVISION HISTORY

| Version | Date       | Author       | Change                                                                                                        |
|---------|------------|--------------|---------------------------------------------------------------------------------------------------------------|
| 0.3     | 2026-10-02 | Amish Chadha | Amish approved the recommendations for open decisions 1 to 3 (FND-DDR-001, O1 to O3); moved to decisions made |
| 0.2     | 2026-10-01 | Amish Chadha | Budget treated as a value-engineering target                                                                  |
| 0.1     | 2026-09-30 | Amish Chadha | Register opened; open decisions moved out of the build plan                                                   |

# FieldNode design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/`; this register is the index Amish works from. The build plan ( `docs/05-build-plan.md` ) describes the design as it stands and does not list open decisions.

## ■ Open decisions

None. All open decisions were decided on 2026-10-02.

## ■ To confirm when parts are bought

| # | What to confirm                                               | Why it matters                                                        | Source      |
|---|---------------------------------------------------------------|-----------------------------------------------------------------------|-------------|
| 1 | The solar panel frame has a flat back lip at least 12 mm wide | The panel clips bolt through it; without it the clip position changes | FND-DDR-003 |
| 2 | The enclosure model, its boss spacing and its lug kit         | They set the internal plate holes and the lug positions               | FND-DDR-003 |
| 3 | The band clamp torque that gives 1,000 N of preload           | The calculation note assumes that preload                             | FND-CAL-001 |

## ■ Value engineering

Value-engineering target: USD 150 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 139 for the base node (USD 11 under the target); USD 148 with the sun shield (USD 2 under the target). Main cost drivers and savings worth trying:

- The largest lines are the power board (USD 22, a prototype carrier price; the board design is TRL 4 work), the enclosure with vent and lug kit (USD 20), the solar panel (USD 14), the controller and LoRa module (USD 14) and the antenna (USD 10).
- Making the design constructable repriced lines 1, 5, 13 and 14 and added line 15, the plug-in connectors and rail fuses (USD 5); the base node rose from USD 126 to USD 139.
- Savings worth trying: the sun shield (USD 9) is an option for hot-climate sites only, so the base node carries none; the power board price should fall once its design is done.

## ■ Decisions made

| Date       | Decision                                                                                                                                                                                                                                | Decided by                                                                                                | Record                   |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|--------------------------|
| 2026-09-25 | TRL 2 review items D1 to D12 (LoRaWAN, STM32WL-class module, one 6 Ah LiFePO4 cell, 6 W panel, two M12 ports, 15 min interval and others)                                                                                               | Amish: "i accept all your recommendations, go with them across all repos."                                | FND-DDR-001, FND-DDR-002 |
| 2026-09-25 | Sun shield for hot-climate sites only; base node mass limit 2.5 kg; 9 V class panel; 100 mW sensor allowance; airtime rule at slow spreading factors                                                                                    | Amish: go with recommendation                                                                             | FND-DDR-002              |
| 2026-09-30 | Design for construction: V-blocks, bottom-face layout, bracket, shield fixing, connector strip, enclosure lugs and other changes that make the node buildable                                                                           | Amish: "i accept your recommended changes on design that are currently being sent across for my approval" | FND-DDR-003              |
| 2026-09-30 | Thumb screws on the shield for the prototype; 5 mm spacers for wall mounting; accept the 0.05 kg mass margin and weigh at TRL 4; fit the shield on the first prototype                                                                  | Amish, same instruction                                                                                   | FND-DDR-003, A1 to A4    |
| 2026-10-02 | Pilot region and radio band: US915 (North America) is the default first variant with a 915 MHz whip; the band switches to that of the first adopting project's site if it is outside North America (EU868, AS923 or IN865)              | Amish: "i approve your recommendations for all 555 open decisions."                                       | FND-DDR-001, O1          |
| 2026-10-02 | Sensor port pin assignment: the candidate pinout of FND-CAL-001 is the proposed standard (pin 1 switched rail, pin 2 data A, pin 3 ground, pin 4 data B, pin 5 analog), sent to HeatMap Node and the next adopting project for sign-off | Amish: "i approve your recommendations for all 555 open decisions."                                       | FND-DDR-001, O2          |
| 2026-10-02 | Firmware update method in the field: by cable inside the box with the lid open (a USB or serial header on the board, no extra hole in the enclosure); over-the-air updates left for a later private-gateway variant                     | Amish: "i approve your recommendations for all 555 open decisions."                                       | FND-DDR-001, O3          |